

$$\int_a^b f(x) dg(x)$$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Chapter 6: The Riemann–Stieltjes Integral

Section: Rectifiable Curves

Based on Walter Rudin's *Principles of Mathematical Analysis*

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Outline

$$\int_a^b f(x) dg(x)$$

$$S(x) = \sum_{i=1}^n f(x_i^*) (g(x_i) - g(x_{i-1})) = S(x_{n-1}^*)$$

Motivation

Curves in $\mathbb{R}^k$

The Length of a Curve

Length as an Integral

Summary



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

$$\int_a^b f(x) dg(x)$$

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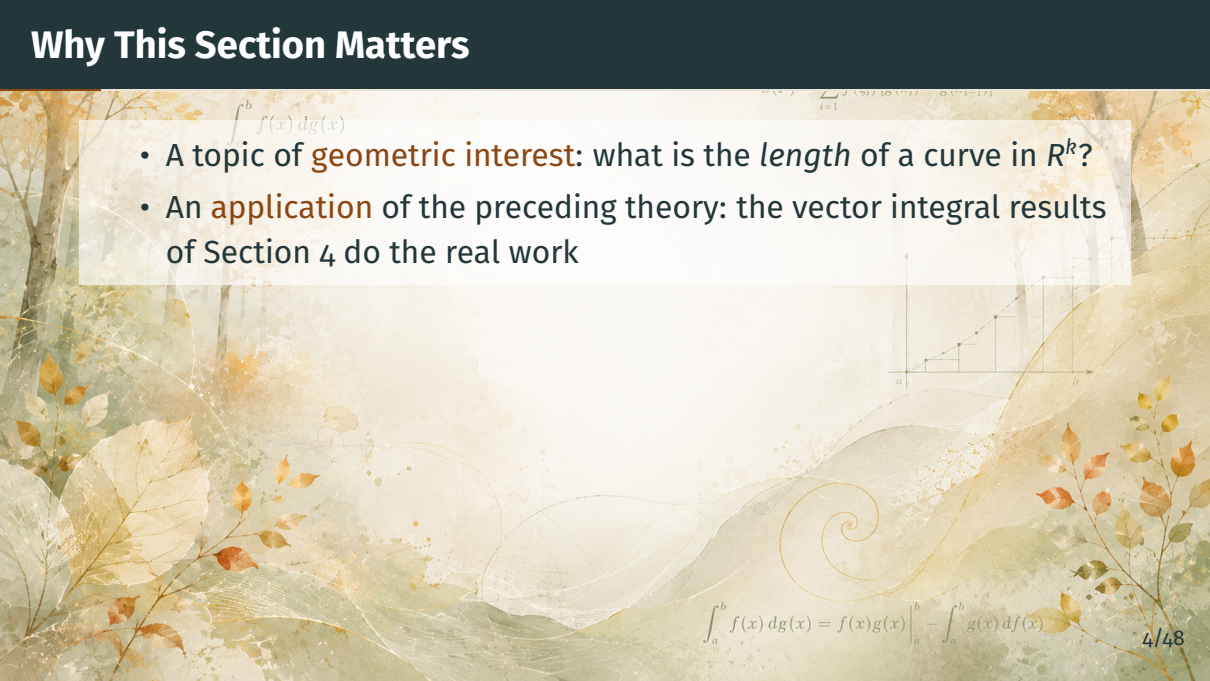
Why This Section Matters

- A topic of **geometric interest**: what is the *length* of a curve in R^k ?



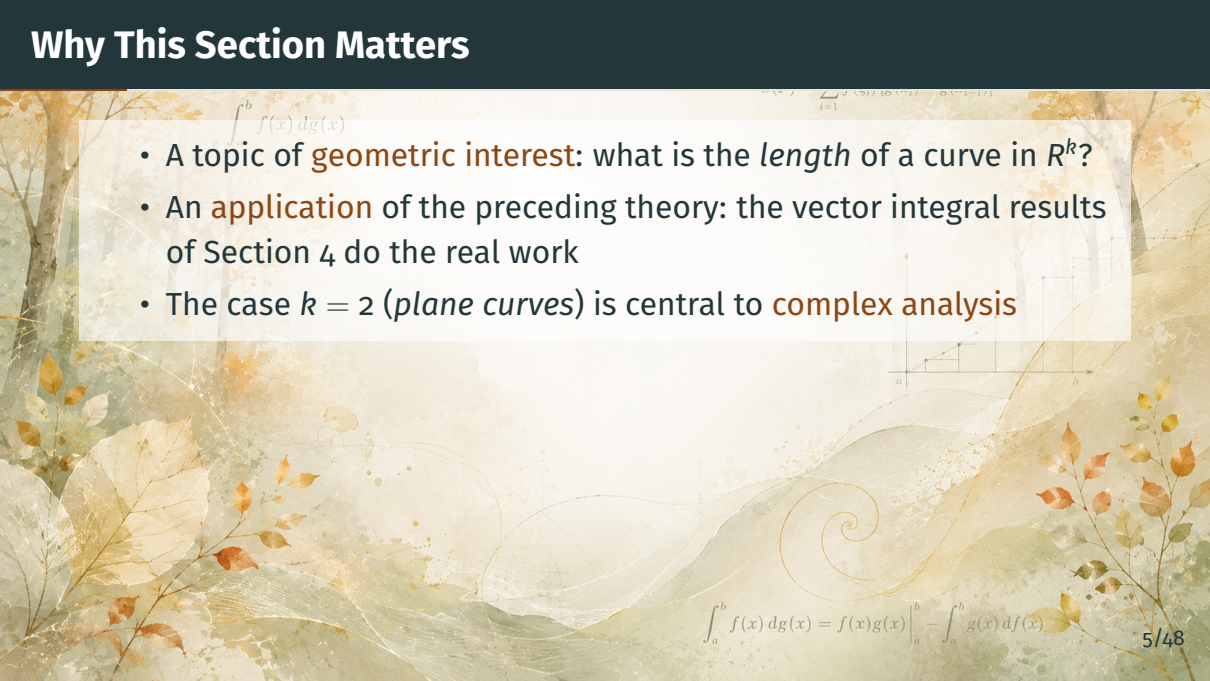
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Why This Section Matters

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- The background of the slide features a soft, painterly illustration of a forest scene with trees and autumn leaves in shades of orange, yellow, and green. Overlaid on this is a semi-transparent white box containing the main text. To the right of the text box, there is a small inset graph showing a curve on a coordinate plane with a step function approximation below it. The x-axis is labeled with 'a' and 'b'. Above the text box, there is a faint mathematical formula: $\sum_{i=1}^n f(x_i^*) (g(x_i) - g(x_{i-1}))$.
- A topic of **geometric interest**: what is the *length* of a curve in R^k ?
 - An **application** of the preceding theory: the vector integral results of Section 4 do the real work

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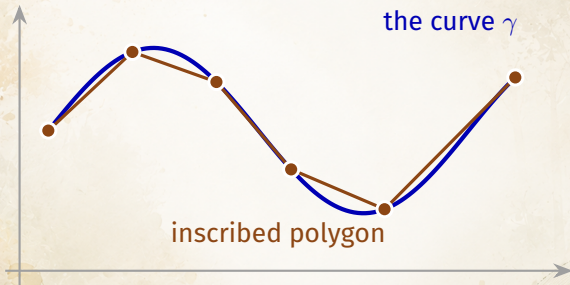
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- A topic of **geometric interest**: what is the *length* of a curve in R^k ?
 - An **application** of the preceding theory: the vector integral results of Section 4 do the real work
 - The case $k = 2$ (*plane curves*) is central to **complex analysis**

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

The Picture to Keep in Mind

$$\int_a^b f(x) dg(x)$$



Length of γ = supremum of the lengths of all inscribed polygons

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

What We Will Prove

- **Definition 6.26:** a *curve* is a continuous map $\gamma : [a, b] \rightarrow \mathbb{R}^k$; *arcs*, *closed curves*;
length $\Lambda(\gamma) = \sup_P \Lambda(P, \gamma)$; *rectifiable* if $\Lambda(\gamma) < \infty$



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- **Theorem 6.27:** if γ' is continuous on $[a, b]$, then γ is rectifiable and

$$\Lambda(\gamma) = \int_a^b |\gamma'(t)| dt$$

length equals the integral of the speed

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

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Curves in R^k



Definition 6.26: Curves, Arcs, Closed Curves

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A **continuous** mapping γ of an interval $[a, b]$ into R^k is called a **curve** in R^k . To emphasize the parameter interval $[a, b]$, we may also say that γ is a curve on $[a, b]$.

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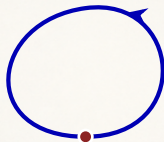
If $\gamma(a) = \gamma(b)$, γ is said to be a **closed curve**.

Three Kinds of Curves

$$\int_a^b f(x) dg(x)$$

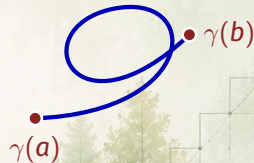


arc
one-to-one



$\gamma(a) = \gamma(b)$
closed curve
returns to its start

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$



neither
crosses itself, not closed

$$a = x_0, x_1, x_2, \dots, x_{n-1}, x_n = b$$

All three are *curves*: continuous maps $\gamma : [a, b] \rightarrow \mathbb{R}^k$.

A Curve Is a Mapping, Not a Point Set

Remark

A curve is defined to be a **mapping**, not a point set. Each curve γ has an associated subset of R^k , its *range* $\gamma([a, b])$, but **different curves may have the same range**.

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Example: same range, different curves



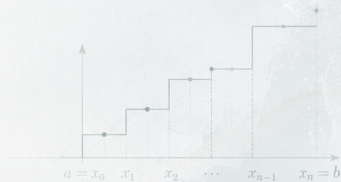
$$\gamma_1(t) = (\cos t, \sin t), 0 \leq t \leq 2\pi$$



$$\gamma_2(t) = (\cos 2t, \sin 2t), 0 \leq t \leq 2\pi$$

$$\int_a^b f(x) dg(x)$$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$



The Length of a Curve

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Inscribed Polygons

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Definition 6.26 (continued): the polygonal length

To each partition $P = \{x_0, \dots, x_n\}$ of $[a, b]$ and each curve γ on $[a, b]$ we associate the number

$$\Lambda(P, \gamma) = \sum_{i=1}^n |\gamma(x_i) - \gamma(x_{i-1})|.$$

$a = x_0 \quad x_1 \quad x_2 \quad \dots \quad x_{n-1} \quad x_n = b$

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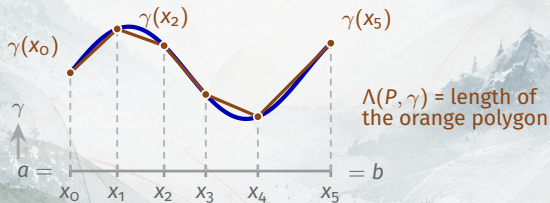
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Refining the Partition

$$\int_a^b f(x) dg(x)$$

$\gamma(x_{i-1})$

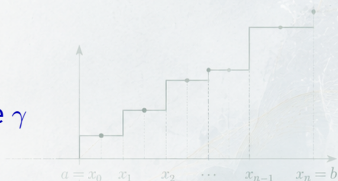
old chord

$\gamma(x_i)$

curve γ

$\gamma(x^*)$ (new point)

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$



Triangle inequality: $|\gamma(x_i) - \gamma(x_{i-1})| \leq |\gamma(x^*) - \gamma(x_{i-1})| + |\gamma(x_i) - \gamma(x^*)|$
 so $P^* \supset P \implies \Lambda(P^*, \gamma) \geq \Lambda(P, \gamma)$: **refining never shortens the polygon**

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Definition 6.26: Length and Rectifiability

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Definition 6.26 (continued): the length of a curve

The **length** of γ is

$$\Lambda(\gamma) = \sup_P \Lambda(P, \gamma),$$

where the supremum is taken over *all* partitions P of $[a, b]$.



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The **length** of γ is

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If $\Lambda(\gamma) < \infty$, we say that γ is **rectifiable**.

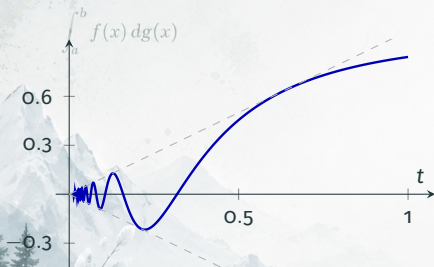
Note: $\Lambda(\gamma)$ always exists in $[0, +\infty]$;
rectifiable means the curve has *finite* length.



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Not Every Curve Is Rectifiable

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$



Example

$\gamma(t) = (t, t \sin(1/t))$ for $0 < t \leq 1$,
 $\gamma(0) = (0, 0)$.

γ is a *continuous* curve in $\mathbb{R}^2$, but
 $\Lambda(\gamma) = +\infty$: it is **not rectifiable**.

Why: the oscillations near $t = 0$ have
heights $\approx \frac{2}{(2m+1)\pi}$, and $\sum \frac{1}{m}$ diverges.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

When Is the Length an Integral?

- In certain cases $\Lambda(\gamma)$ is given by a **Riemann integral**
- We prove this for **continuously differentiable** curves: γ' exists and is continuous on $[a, b]$

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- In certain cases $\Lambda(\gamma)$ is given by a **Riemann integral**
- We prove this for **continuously differentiable** curves: γ' exists and is continuous on $[a, b]$
- **Heuristic:** $\gamma'(t)$ = velocity, $|\gamma'(t)|$ = speed, so over a short time Δx_i :

$$|\gamma(x_i) - \gamma(x_{i-1})| \approx |\gamma'(x_i)| \Delta x_i,$$

$$\Lambda(P, \gamma) \approx \sum_i |\gamma'(x_i)| \Delta x_i \longrightarrow \int_a^b |\gamma'(t)| dt$$

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Length as an Integral



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Theorem 6.27: Length of a Continuously Differentiable Curve

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If γ' is continuous on $[a, b]$, then γ is rectifiable, and

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

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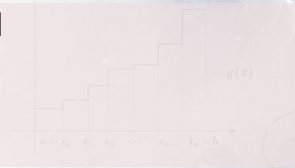
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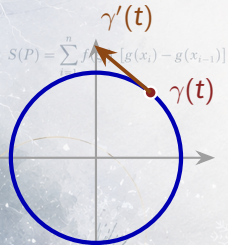
Key idea: distance traveled = integral of speed.

In coordinates, $\gamma = (\gamma_1, \dots, \gamma_k)$ and $\Lambda(\gamma) = \int_a^b \sqrt{\gamma_1'(t)^2 + \dots + \gamma_k'(t)^2} dt$, the familiar *arc length formula* from calculus.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Example: The Unit Circle

$$\int_a^b f(x) dg(x)$$



Example: $\gamma(t) = (\cos t, \sin t)$, $0 \leq t \leq 2\pi$

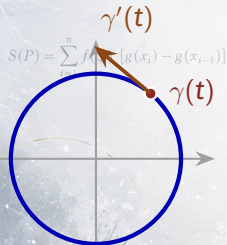
$\gamma'(t) = (-\sin t, \cos t)$ is continuous,
and $|\gamma'(t)| = \sqrt{\sin^2 t + \cos^2 t} = 1$.

By Theorem 6.27: $\Lambda(\gamma) = \int_0^{2\pi} 1 dt = 2\pi$.

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By Theorem 6.27: $\Lambda(\gamma) = \int_0^{2\pi} 1 dt = 2\pi$.

Same range, twice around

$\gamma_2(t) = (\cos 2t, \sin 2t)$, $0 \leq t \leq 2\pi$: $|\gamma_2'(t)| = 2$,
so $\Lambda(\gamma_2) = \int_0^{2\pi} 2 dt = 4\pi$.

Length depends on the **mapping**, not on the range.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Recall: Two Tools from Section 4

Recall: Theorem 6.24 (Fundamental Theorem, vector version)

If $\mathbf{f} \in \mathcal{R}$ on $[a, b]$ and $\mathbf{F}' = \mathbf{f}$ on $[a, b]$, then $\int_a^b \mathbf{f}(t) dt = \mathbf{F}(b) - \mathbf{F}(a)$.

Applied to $\mathbf{F} = \gamma$ on a subinterval $[x_{i-1}, x_i]$: $\gamma(x_i) - \gamma(x_{i-1}) = \int_{x_{i-1}}^{x_i} \gamma'(t) dt$.

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Recall: Theorem 6.25 (triangle inequality for integrals)

If $\mathbf{f} \in \mathcal{R}(\alpha)$ on $[a, b]$, then $|\mathbf{f}| \in \mathcal{R}(\alpha)$ and $\left| \int_a^b \mathbf{f} d\alpha \right| \leq \int_a^b |\mathbf{f}| d\alpha$.

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Proof Strategy: Theorem 6.27

$$\int_a^b f(x) dg(x)$$

Goal: $\Lambda(\gamma) = \int_a^b |\gamma'(t)| dt$ (and in particular $\Lambda(\gamma) < \infty$)

Plan: prove **two inequalities**.

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($\leq$) Each chord is at most the integral of the speed over its subinterval (Theorems 6.24 and 6.25); sum over i ; take the sup over P .



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Plan: prove **two inequalities**.

- ($\leq$) Each chord is at most the integral of the speed over its subinterval (Theorems 6.24 and 6.25); sum over i ; take the sup over P .
- ($\geq$) Fix $\varepsilon > 0$. **Uniform continuity** of γ' gives a fine partition on which γ' is nearly constant, so the integral of the speed over each subinterval is at most the chord plus $2\varepsilon \Delta x_i$. Sum, then let $\varepsilon \rightarrow 0$.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof — Part 1: $\Lambda(\gamma) \leq \int_a^b |\gamma'(t)| dt$

Step 1: bound one chord

If $a \leq x_{i-1} < x_i \leq b$, then by Theorems 6.24 and 6.25,

$$|\gamma(x_i) - \gamma(x_{i-1})| = \left| \int_{x_{i-1}}^{x_i} \gamma'(t) dt \right| \leq \int_{x_{i-1}}^{x_i} |\gamma'(t)| dt.$$



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Step 2: sum over the subintervals

Hence, for every partition P of $[a, b]$,

$$\Lambda(P, \gamma) = \sum_i |\gamma(x_i) - \gamma(x_{i-1})| \leq \sum_i \int_{x_{i-1}}^{x_i} |\gamma'(t)| dt = \int_a^b |\gamma'(t)| dt.$$

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$$\Lambda(P, \gamma) = \sum_i |\gamma(x_i) - \gamma(x_{i-1})| \leq \sum_i \int_{x_{i-1}}^{x_i} |\gamma'(t)| dt = \int_a^b |\gamma'(t)| dt.$$

Step 3: take the supremum over P

Sup over P : $\Lambda(\gamma) \leq \int_a^b |\gamma'(t)| dt < \infty$, so γ is **rectifiable**.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof — Part 2: Setup

Step 1: uniform continuity of γ'

Let $\varepsilon > 0$. Since γ' is continuous on the compact set $[a, b]$, it is **uniformly continuous** (Theorem 4.19): there is $\delta > 0$ such that $|\gamma'(s) - \gamma'(t)| < \varepsilon$ whenever $|s - t| < \delta$.

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Step 2: choose a fine partition

Let $P = \{x_0, \dots, x_n\}$ be a partition of $[a, b]$ with $\Delta x_i < \delta$ for all i .

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Let $P = \{x_0, \dots, x_n\}$ be a partition of $[a, b]$ with $\Delta x_i < \delta$ for all i .

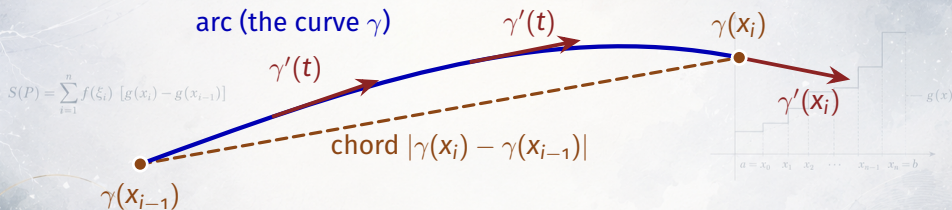
Step 3: the speed is nearly constant on each piece

If $x_{i-1} \leq t \leq x_i$, then $|t - x_i| \leq \Delta x_i < \delta$, so $|\gamma'(t)| \leq |\gamma'(x_i)| + |\gamma'(t) - \gamma'(x_i)| < |\gamma'(x_i)| + \varepsilon$.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof — Part 2: Why a Fine Partition Helps

$$\int_a^b f(x) dg(x)$$



velocities nearly equal: $|\gamma'(t) - \gamma'(x_i)| < \varepsilon$ for all $t \in [x_{i-1}, x_i]$,
so chord $\approx$ arc $\approx |\gamma'(x_i)| \Delta x_i$

Goal for one piece: $\int_{x_{i-1}}^{x_i} |\gamma'(t)| dt \leq |\gamma(x_i) - \gamma(x_{i-1})| + 2\varepsilon \Delta x_i$

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof — Part 2: The Key Estimate on One Piece

$$\int_a^b f(x) dg(x)$$

Step 4: for the subinterval $[x_{i-1}, x_i]$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})] \leq \int_{x_{i-1}}^{x_i} |\gamma'(t)| dt \leq |\gamma'(x_i)| \Delta x_i + \varepsilon \Delta x_i$$



Integrate the bound $|\gamma'(t)| \leq |\gamma'(x_i)| + \varepsilon$ from Step 3 over $[x_{i-1}, x_i]$.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof — Part 2: The Key Estimate on One Piece

$$\int_a^b f(x) dg(x)$$

Step 4: for the subinterval $[x_{i-1}, x_i]$

$$S(P) = \sum_{i=1}^n f(\xi_i) \int_{x_{i-1}}^{x_i} g(x_i) |\gamma'(t)| dt \leq |\gamma'(x_i)| \Delta x_i + \varepsilon \Delta x_i$$

$$= \left| \int_{x_{i-1}}^{x_i} [\gamma'(t) + \gamma'(x_i) - \gamma'(t)] dt \right| + \varepsilon \Delta x_i$$



$\gamma'(x_i)$ is a *constant vector*, so $|\gamma'(x_i)| \Delta x_i = \left| \int_{x_{i-1}}^{x_i} \gamma'(x_i) dt \right|$;
then **add and subtract** $\gamma'(t)$ inside.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof — Part 2: The Key Estimate on One Piece

Step 4: for the subinterval $[x_{i-1}, x_i]$

$$\int_{x_{i-1}}^{x_i} |\gamma'(t)| dt \leq |\gamma'(x_i)| \Delta x_i + \varepsilon \Delta x_i$$

$$= \left| \int_{x_{i-1}}^{x_i} [\gamma'(t) + \gamma'(x_i) - \gamma'(t)] dt \right| + \varepsilon \Delta x_i$$

$$\leq \left| \int_{x_{i-1}}^{x_i} \gamma'(t) dt \right| + \left| \int_{x_{i-1}}^{x_i} [\gamma'(x_i) - \gamma'(t)] dt \right| + \varepsilon \Delta x_i$$

Split the integral and use the **triangle inequality** $|\mathbf{u} + \mathbf{v}| \leq |\mathbf{u}| + |\mathbf{v}|$.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof — Part 2: The Key Estimate on One Piece

Step 4: for the subinterval $[x_{i-1}, x_i]$

$$\begin{aligned} \int_{x_{i-1}}^{x_i} |\gamma'(t)| dt &\leq |\gamma'(x_i)| \Delta x_i + \varepsilon \Delta x_i \\ &= \left| \int_{x_{i-1}}^{x_i} [\gamma'(t) + \gamma'(x_i) - \gamma'(t)] dt \right| + \varepsilon \Delta x_i \\ &\leq \left| \int_{x_{i-1}}^{x_i} \gamma'(t) dt \right| + \left| \int_{x_{i-1}}^{x_i} [\gamma'(x_i) - \gamma'(t)] dt \right| + \varepsilon \Delta x_i \\ &\leq |\gamma(x_i) - \gamma(x_{i-1})| + 2\varepsilon \Delta x_i. \end{aligned}$$

First term: the **chord** (Thm 6.24). Second: $\leq \varepsilon \Delta x_i$ (Thm 6.25 and Step 1).

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof — Part 2: Adding Up

Step 5: sum over $i = 1, \dots, n$

$$\begin{aligned} \int_a^b |\gamma'(t)| dt &= \sum_{i=1}^n \int_{x_{i-1}}^{x_i} |\gamma'(t)| dt \leq \sum_{i=1}^n |\gamma(x_i) - \gamma(x_{i-1})| + 2\varepsilon \sum_{i=1}^n \Delta x_i \\ &= \Lambda(P, \gamma) + 2\varepsilon (b - a) \end{aligned}$$

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$\Lambda(P, \gamma) \leq \Lambda(\gamma)$ because $\Lambda(\gamma)$ is the supremum over all partitions.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof — Part 2: Adding Up

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Step 6: let $\varepsilon \rightarrow 0$

ε was arbitrary: $\boxed{\int_a^b |\gamma'(t)| dt \leq \Lambda(\gamma)}$. With Part 1: $\Lambda(\gamma) = \int_a^b |\gamma'(t)| dt$. ■

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

$$\int_a^b f(x) dg(x)$$

Summary



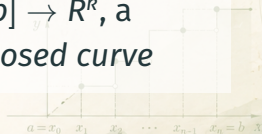
$$\sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Summary

$$\int_a^b f(x) dg(x)$$

Key takeaways from this section:

- **Definition 6.26:** a **curve** is a continuous map $\gamma : [a, b] \rightarrow \mathbb{R}^k$, a mapping, not a set; one-to-one: *arc*; $\gamma(a) = \gamma(b)$: *closed curve*



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Summary

$$\int_a^b f(x) dg(x)$$

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- **Length:** $\Lambda(\gamma) = \sup_P \Lambda(P, \gamma)$, the sup of **inscribed polygon** lengths; *rectifiable* means $\Lambda(\gamma) < \infty$ (not automatic!)

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Summary

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- **Definition 6.26:** a **curve** is a continuous map $\gamma : [a, b] \rightarrow R^k$, a mapping, not a set; one-to-one: *arc*; $\gamma(a) = \gamma(b)$: *closed curve*
- **Length:** $\Lambda(\gamma) = \sup_P \Lambda(P, \gamma)$, the sup of **inscribed polygon** lengths; *rectifiable* means $\Lambda(\gamma) < \infty$ (not automatic!)
- **Theorem 6.27:** γ' continuous $\Rightarrow$ rectifiable, $\Lambda(\gamma) = \int_a^b |\gamma'(t)| dt$ (*length = integral of speed*); proof: Theorems 6.24, 6.25 + **uniform continuity**

Coming up next: Chapter 7, **Sequences and Series of Functions.**